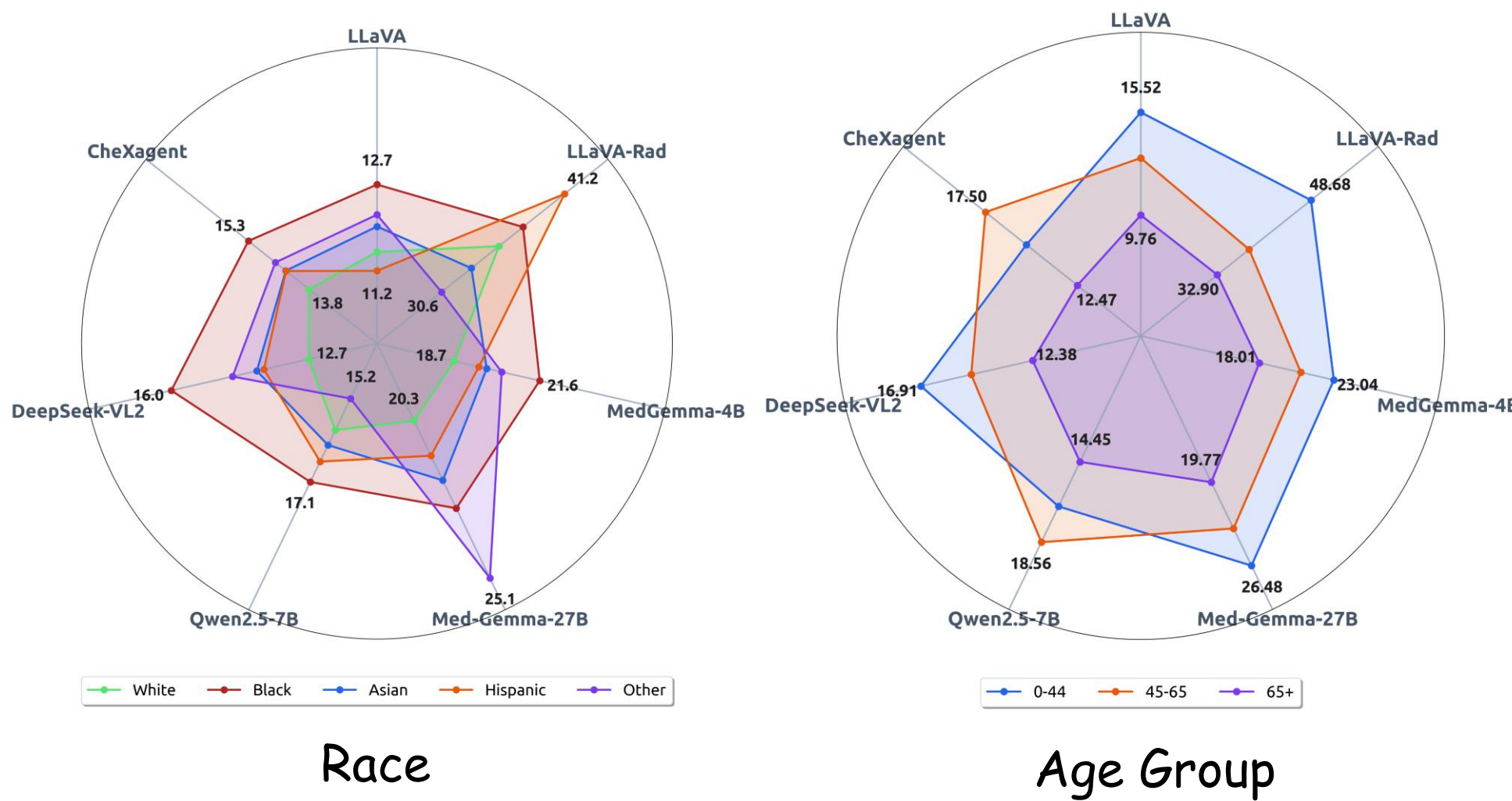
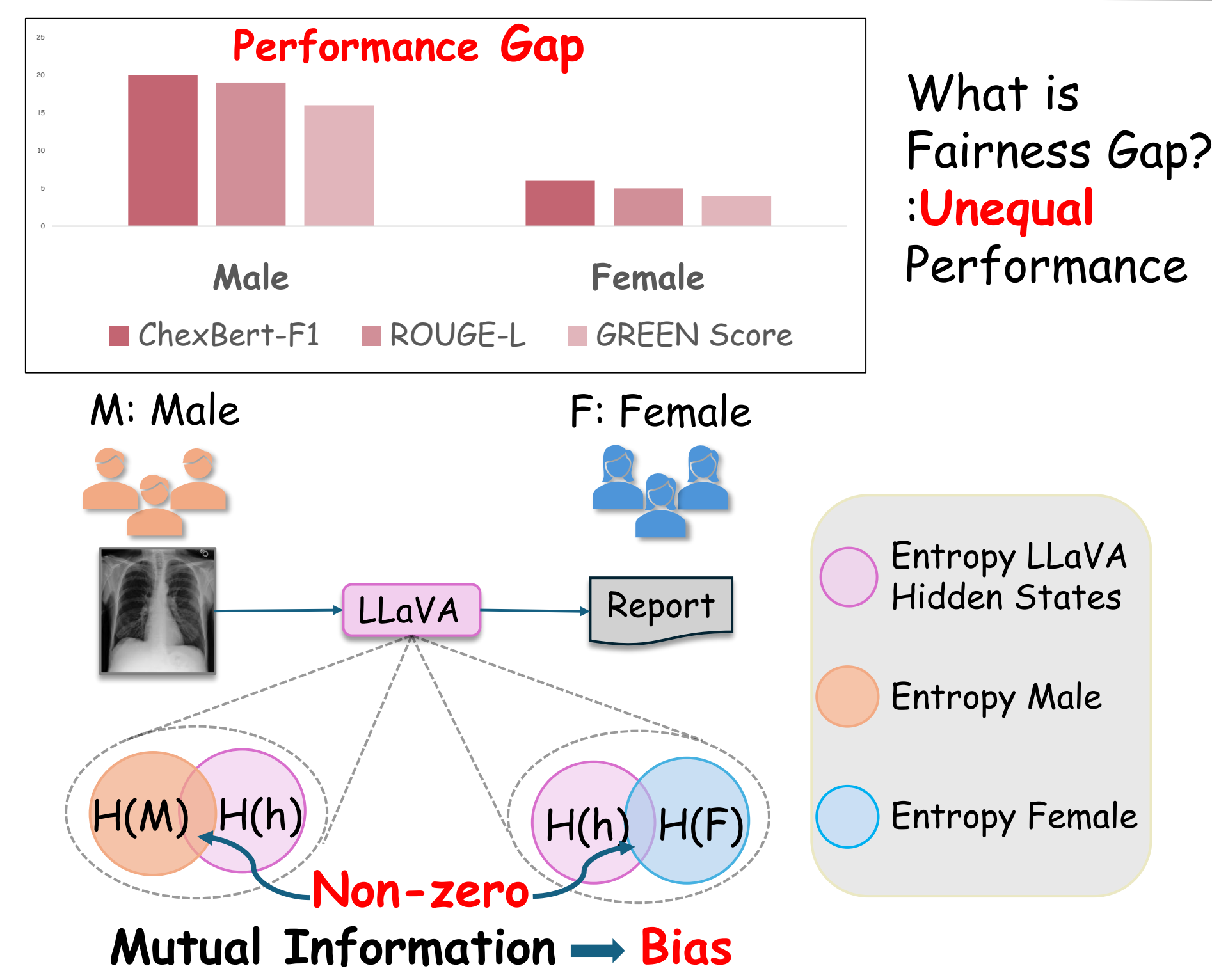


Is Your Vision Language Model Fair?



No, multiple opensource Multi-Modal LLMs show fairness gap across demographic attributes.



How to Measure Fairness ?

- Tradeoff** : Fairness should not come from making every group perform poorly.
- Fairness gap**: best group - worst group.
 $\Delta M_a = \max_{a \in Z_a} M_a - \min_{a \in Z_a} M_a$
- Use **Equity Scaled** Metric: $\uparrow M_{\text{all}}$ and $\downarrow M_a$
$$ES-M_a = \frac{M_{\text{all}}}{1 + \Delta M_a}$$

FairLLaVA: Fairness-Aware Parameter-Efficient Fine-Tuning for Large Vision-Language Assistants.

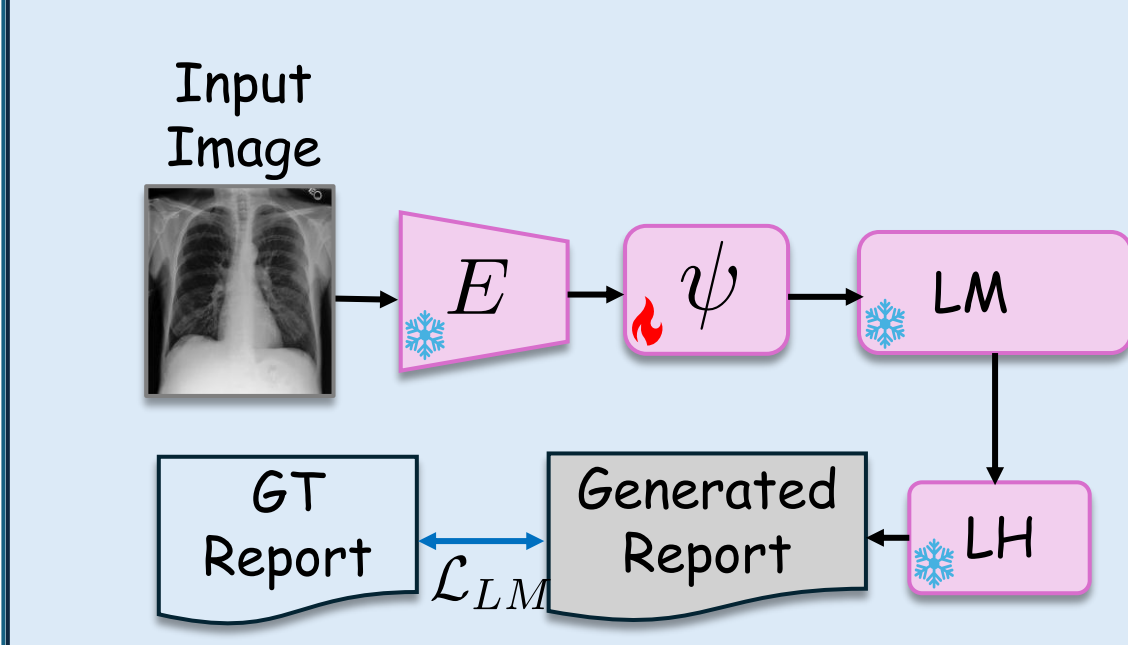
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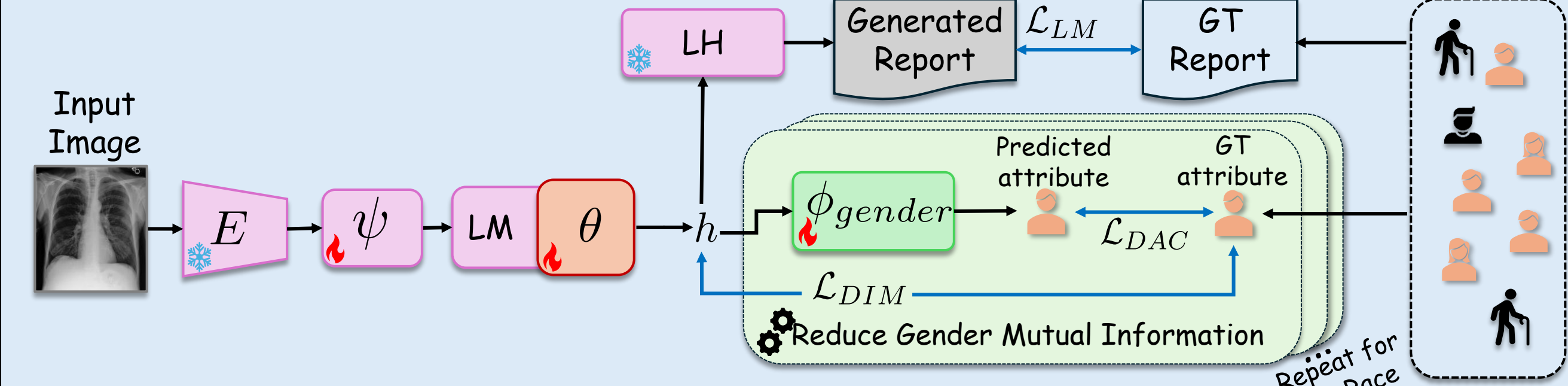


Solution: Reduce Mutual Information (MI) to Reduce Spurious Correlations.

Stage 1: Visual Language Alignment



Stage 2: Fairness-Aware Finetuning Via Mutual Information Minimization



E : Image Embedder ψ : Multimodal Projector LM : Language Model ϕ : Demographic Attribute Classifier L_{DIM} : Demography Information Minimization
 LH : Language Model Head h : Hidden State θ : LoRA Adapters L_{LM} : Language Model Loss L_{DAC} : Demographic Attribute Classifier Loss

Results: FairLLaVA reduces Fairness Gap.

Method	Race↓			Age Group↓			Gender↓			Overall↑		
	BLEU-4	RadGraph-F1	GREEN	BLEU-4	RadGraph-F1	GREEN	BLEU-4	RadGraph-F1	GREEN	BLEU-4	RadGraph-F1	GREEN
LLaVA-Rad	6.20	6.20	8.02	3.39	19.97	16.91	0.23	2.23	2.29	15.40	29.80	35.82
MedGemma-4B	0.69	4.70	2.96	0.39	5.98	5.61	0.04	0.45	1.67	1.61	10.54	20.02
MedGemma-27B	0.06	4.13	4.73	0.56	8.33	5.70	0.00	1.18	1.51	1.82	14.46	22.45
Qwen2.5-7B	0.22	2.66	1.88	0.31	5.44	3.92	0.04	0.23	0.32	1.80	10.18	16.098
DeepSeek-VL2	0.21	2.73	3.34	0.44	3.90	4.64	0.07	0.31	0.51	1.78	8.12	14.16
Reweighting-All	3.44	5.27	5.50	1.40	8.58	14.93	0.11	1.10	0.04	7.69	20.76	24.38
Resampling-All	4.90	7.66	15.25	3.03	7.49	16.05	0.23	0.60	1.15	13.69	25.55	34.61
Adv. MLP Classifier-All	0.74	3.51	3.00	0.70	1.63	11.32	0.20	0.56	0.31	1.88	10.61	9.36
FairLLaVA-All	0.61	3.50	9.97	1.01	5.60	12.29	0.45	0.47	1.80	13.92	28.52	34.32

Results: FairLLaVA improves Equity Scaled Metrics.

Method	Race↑				Age Group↑				Gender↑			
	ES-BLEU-1	ES-BLEU-4	ES-RadGraph-F1	ES-GREEN	ES-BLEU-1	ES-BLEU-4	ES-RadGraph-F1	ES-GREEN	ES-BLEU-1	ES-BLEU-4	ES-RadGraph-F1	ES-GREEN
LLaVA	2.90	1.12	1.63	2.47	2.61	0.81	0.95	1.36	8.42	0.87	5.19	6.73
LLaVA-Rad	5.29	2.14	4.14	3.97	8.28	3.51	1.42	2.00	28.06	12.53	9.24	10.88
MedGemma-4B	2.36	0.95	1.85	5.05	3.17	1.16	1.51	3.03	14.18	1.55	7.27	7.51
MedGemma-27B	2.01	1.72	2.82	3.92	3.98	1.17	1.55	3.35	17.80	1.82	6.64	8.92
Qwen2.5-7B	7.52	1.47	2.78	5.59	6.16	1.37	1.58	3.27	16.92	1.73	8.28	12.27
DeepSeek-VL2	4.09	1.47	2.18	3.27	3.00	1.23	1.66	2.51	11.35	1.66	6.20	9.37
CheXagent	4.70	1.05	1.73	1.49	2.21	0.73	0.89	1.37	9.11	1.04	4.17	3.46
Reweighting-All	1.81	1.73	3.31	3.75	7.36	3.20	2.17	1.53	11.72	6.92	9.88	23.42
Resampling-All	8.71	2.32	2.95	2.13	12.54	3.40	3.01	2.03	30.38	11.13	15.97	16.11
Adv. MLP Classifier-All	3.92	1.08	2.35	2.34	6.74	1.11	4.03	0.76	14.89	1.57	6.80	7.15
FairLLaVA-All	13.36	8.65	6.34	3.11	21.89	6.93	4.06	2.78	24.89	9.60	19.40	12.27

Mutual Information between $\mathbf{a}$ and h_θ^l :

$$\mathcal{I}(\mathbf{a}, h_\theta^l(x) | (x, \mathbf{a}) \sim \mathcal{D}) = \mathbb{E}_{(x, \mathbf{a}) \sim \mathcal{D}} \left[\log \frac{P_{\mathcal{D}}(\mathbf{a}, h_\theta^l(x))}{P_{\mathcal{D}}(\mathbf{a}) P_{\mathcal{D}}(h_\theta^l(x))} \right]$$

But MI is intractable ☹, Use tractable upper bound.

$$\mathcal{I}^u(\mathbf{a}, h^l(x)) = \mathbb{E}_{(\mathbf{a}, x) \sim \mathcal{D}} [\log \phi(\mathbf{a} | h^l(x))] - \mathbb{E}_{(\mathbf{a}, x) \sim \mathcal{D}} [\log \phi(\mathbf{a} | h^l(x'))] \quad x^{i.i.d.} \sim \mathcal{D}$$

Where ϕ is an attribute classifier, trained with,

$$\mathcal{L}_{DAC}(\phi) = -\mathbb{E}_{(\mathbf{a}, x) \sim \mathcal{D}} [\log \phi(\mathbf{a} | h(x))]$$

Hispanic

White

Ground Truth: Cardiomegaly

Ground Truth Report	The lungs are clear of large confluent consolidation. Cardiac silhouette enlarged but could be accentuated by positioning and relatively low inspiratory effort. Calcifications noted at the aortic arch. Degenerative changes noted at the glenohumeral joints bilaterally. Osseous and soft tissue structures otherwise unremarkable.	There is prominence of the interstitial markings suggesting moderate interstitial edema. No large pleural effusion is seen. There is no evidence of pneumothorax. The cardiac silhouette is enlarged . The aorta is tortuous.
LLaVA-Rad Report	The cardiac silhouette remains enlarged . The aortic knob is calcified. There are relatively low lung volumes. No definite focal consolidation is seen. There is no large pleural effusion or pneumothorax. No overt pulmonary edema is seen.	Lower lung volumes seen on the current exam. The lungs are clear of consolidation or effusion. The cardiomediastinal silhouette looks normal given differences in positioning and technique. No acute osseous abnormality is identified.
FaiLLaVA Report	The lungs are clear of focal consolidation or large effusion. Cardiac silhouette is enlarged but stable. Atherosclerotic calcifications noted at the aortic arch. No acute osseous abnormality is identified.	There are relatively low lung volumes. The cardiac silhouette is enlarged . There is mild pulmonary vascular congestion. No definite focal consolidation is seen. There is no pleural effusion or pneumothorax.